3 Image Enhancement in the Spatial Domain

- Image Enhancement:
  - The objective of image enhancement is to process an image so that the result is more suitable than the original image for a specific application.
- Reasons:
  - Contrast enhancement
  - Visual improvement
  - Image understanding

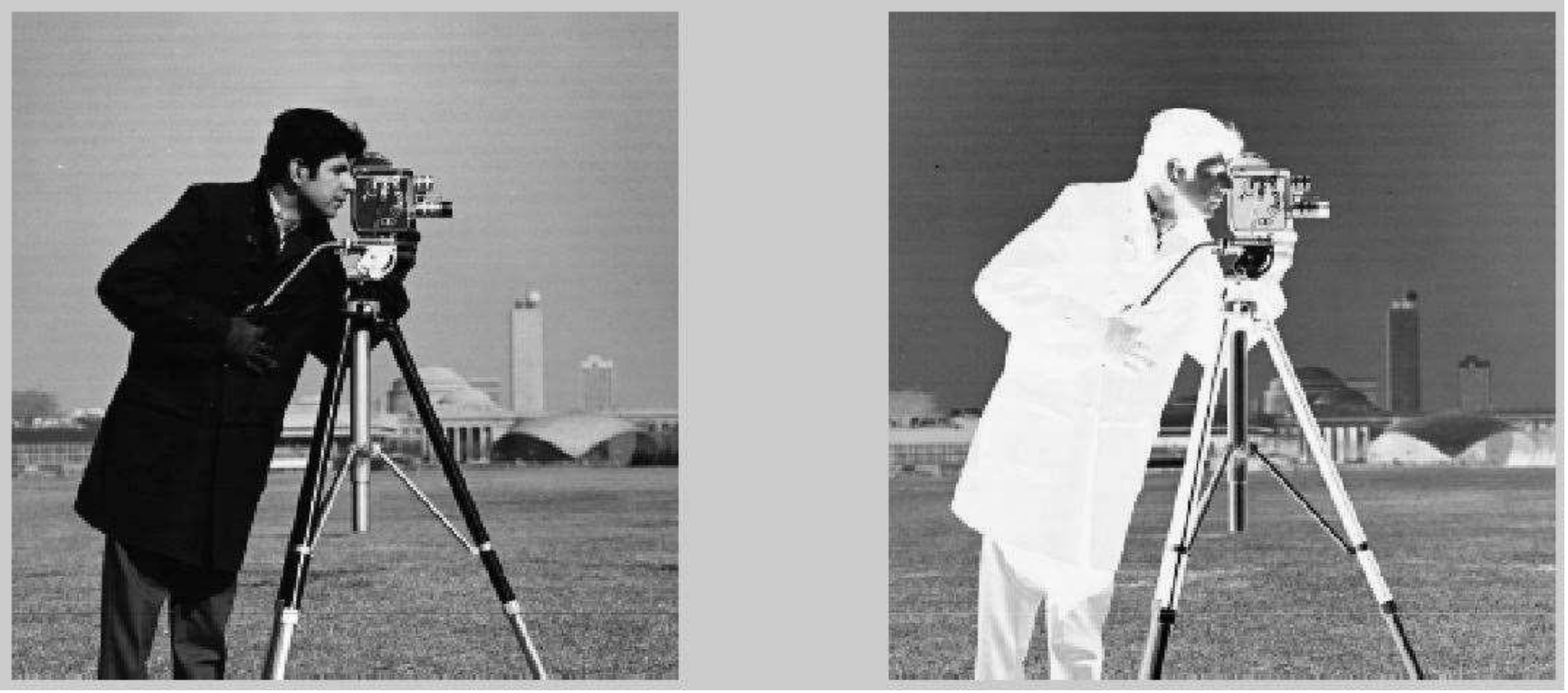
Enhancement Methods

- Spatial Domain
  - Based on direct manipulation of pixels in an image
- Frequency Domain
  - Based on modifying the Fourier transform of an image

Image Negatives

- $s = L - 1 - r$
  - s is the output intensity value
  - L is the highest intensity levels
  - r is the input intensity value
- Particularly suited for enhancing white or gray detail embedded in dark regions of an image, especially when the black areas are dominant in size.
- Produces photographic negative.
- Some details are easier to spot if we go from black and white to white and black.

Image Negative



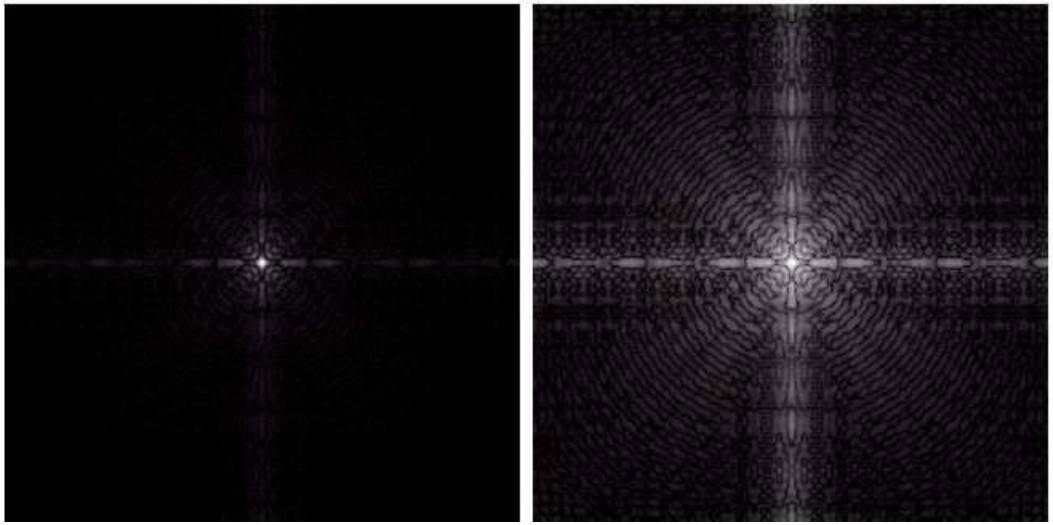
Log Transformations

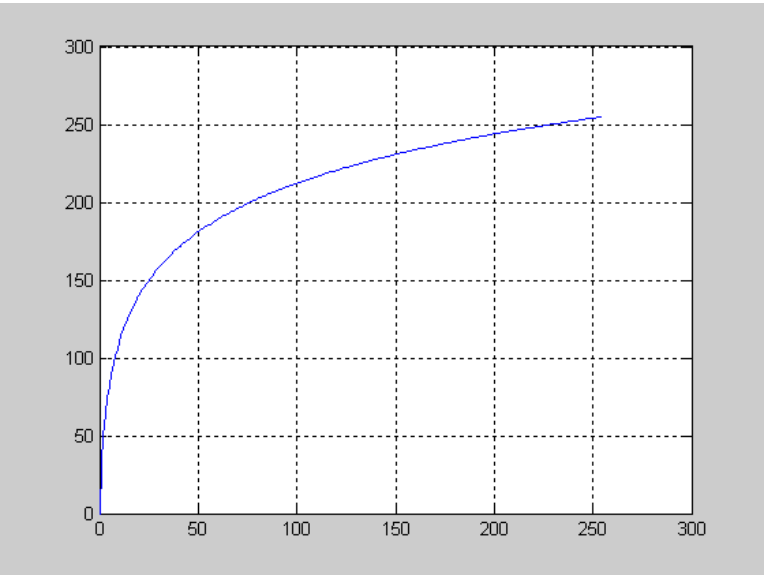
- $s = c\log(1 + r)$, c is constant.
- It maps a narrow range of low intensity values in the input into a wide range of output levels.
- The opposite is true of higher values of input levels.
- It expands the values of dark pixels in an image while compressing the higher level values.
- It compresses the dynamic range of images with large variations in pixel values.
- The opposite is true of the inverse log transformation.

Log Transformations

a b

FIGURE 3.5
(a) Fourier spectrum.
(b) Result of applying the log transformation given in Eq. (3.2-2) with $c = 1$.





$$T(r) = c\log(1+r)$$



Power Law (Gamma) Transformations

- $s = cr^{\gamma}$, c and γ are both positive constants.
- With fractional values($0<\gamma<1$) of gamma map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values ($\gamma > 1$)of input levels.
- Variety of devices used for image capture, printing, and display respond according to a power law.
- Process used to correct these power law response phenomena is called *gamma correction.*

Power Law (Gamma) Transformations

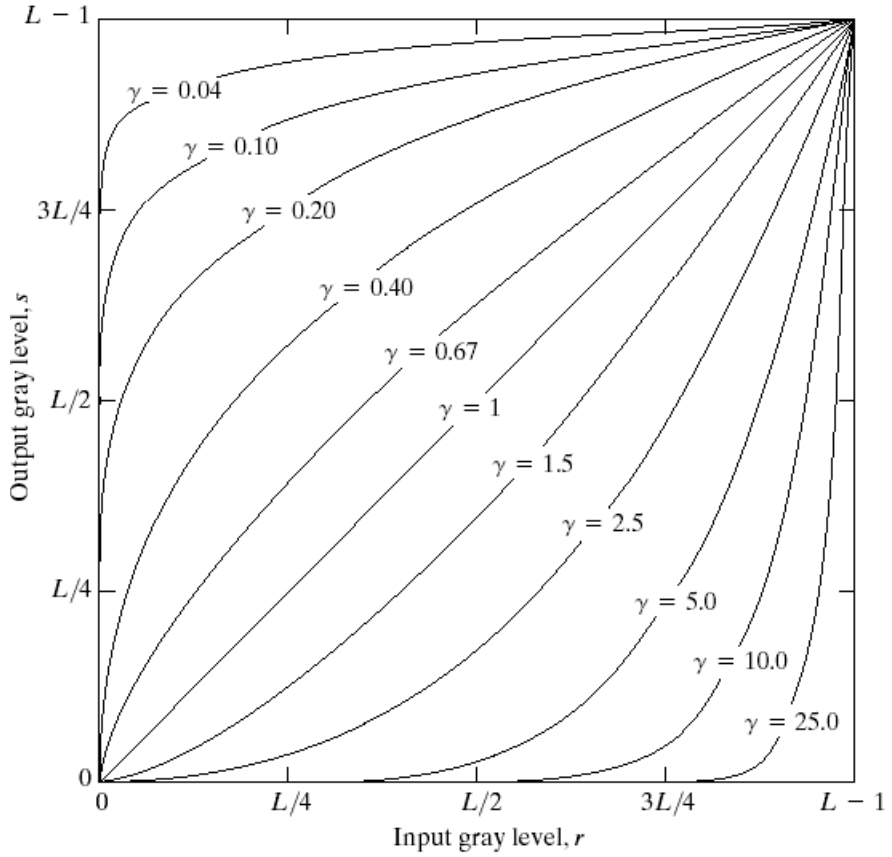
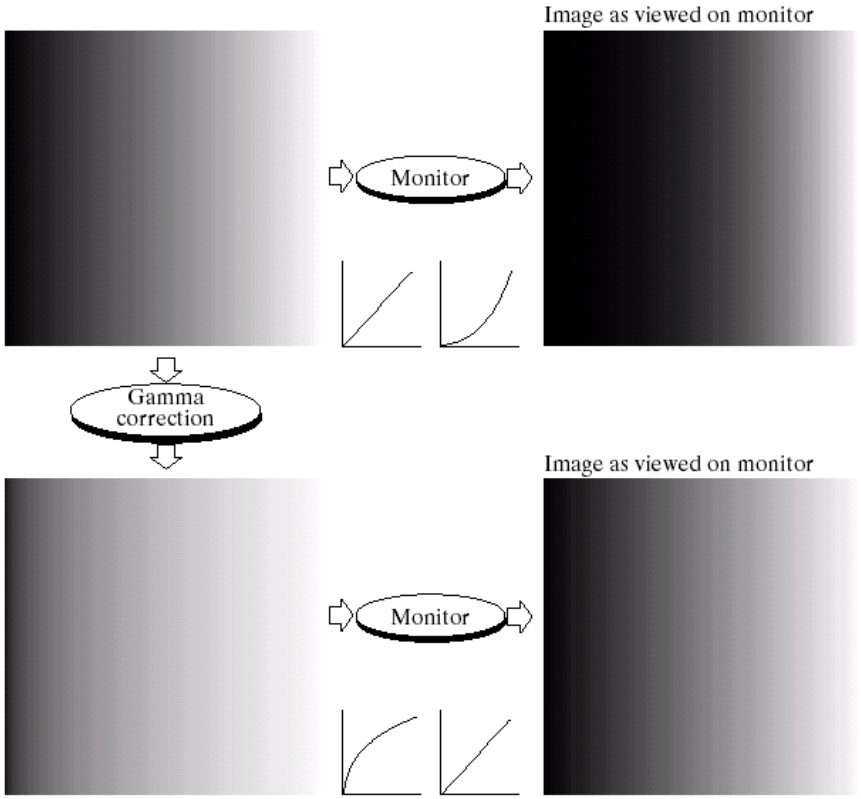


FIGURE 3.6 Plots of the equation $s = cr^{\gamma}$ for various values of γ ($c = 1$ in all cases).

Power Law (Gamma) Transformations

- CRT devices have Intensity-voltage response functions that are power functions.
- They vary in exponents from 1.8 to 2.5.



a b
c d

FIGURE 3.7
(a) Linear-wedge gray-scale image. (b) Response of monitor to linear wedge. (c) Gamma-corrected wedge. (d) Output of monitor.

$$T(r) = r^{1/2.5} = r^{0.4}$$

Histogram Processing

- Histogram of a digital image with intensity levels in the range [0,L-1] is a discrete function $h(r_k) = n_k$, where r_k is the kth intensity value and n_k is the number of pixels in the image with intensity r_k.
- Normalized histogram $p(r_k)=n_k/MN$, for k = 0,1,2..... L-1. Spatial image resolution is M×N.
- Histogram manipulation can be used for image enhancement.
- Information inherent in histogram is also quite useful in other image processing applications, such as image compression and segmentation.

Histogram



Image

0	7	3	2	3
0	0	0	6	7
7	7	2	2	0
1	1	0	4	1
0	0	7	4	1

Image matrix

Similarly

INTENSITY r	0	1	2	3	4	5	6	7
NUMBER of pixels of r h(r)	h(r0)=**8**	h(r1)=**4**	h(r2)=**3**	h(r3)=**2**	h(r4)=**2**	h(r5)=**0**	h(r6)=**1**	h(r7)=**5**

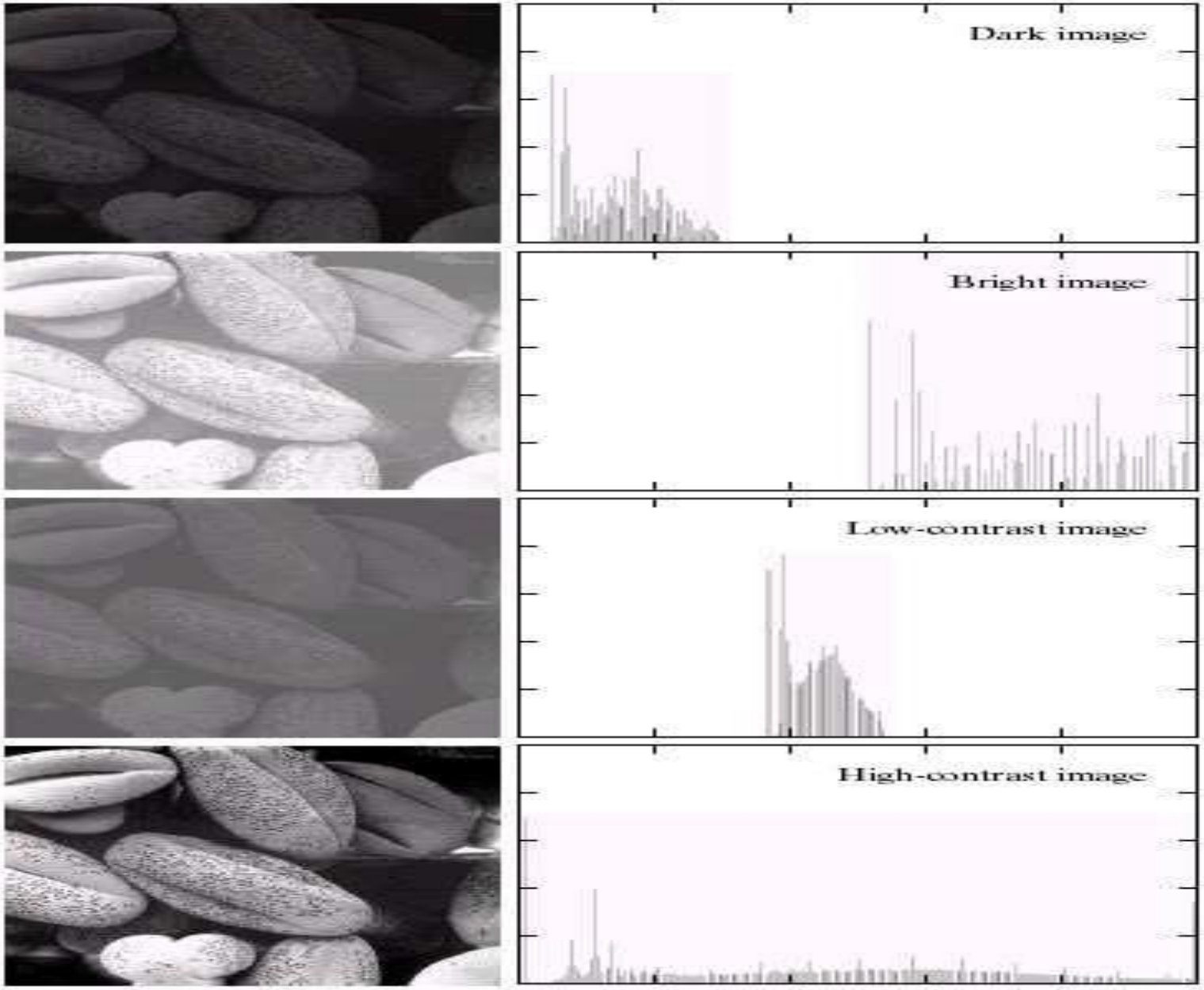
Histogram

r	0	1	2	3	4	5	6	7
h(r)	8	4	3	2	2	0	1	5
p(r) h(r)/(5*5)	8/25	4/25	3/25	2/25	2/25	0/25	1/25	5/25

HISTOGRAM - ***h(r)*** - Y axis - ***number*** of intensities

NORMALIZED *HISTOGRAM* - ***p(r)*** - Y axis - ***probability*** of intensities

Histogram Analysis



Histogram Equalization

- Intensity mapping form

$$s = T(r),\ 0 \le r \le L-1$$

Conditions:

a) T(r) is a monotonically increasing function in the interval [0, L-1] and

b) $0 \le T(r) \le L-1 \qquad for\ 0 \le r \le L-1$

In some formulations, we use the inverse in which case

$$r = T^{-1}(s),\ 0 \le s \le 1$$

(a) change to

a´) T(r) is a strictly monotonically increasing function in the interval [0, L-1]

Histogram Equalization

- Intensity levels in an image may be viewed as random variables in the interval [0,L-1]
- Fundamental descriptor of a random variable is its probability density function (PDF).
- Let $p_r(r)$ and $p_s(s)$ denote the PDFs of r and s Respectively.

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right|$$

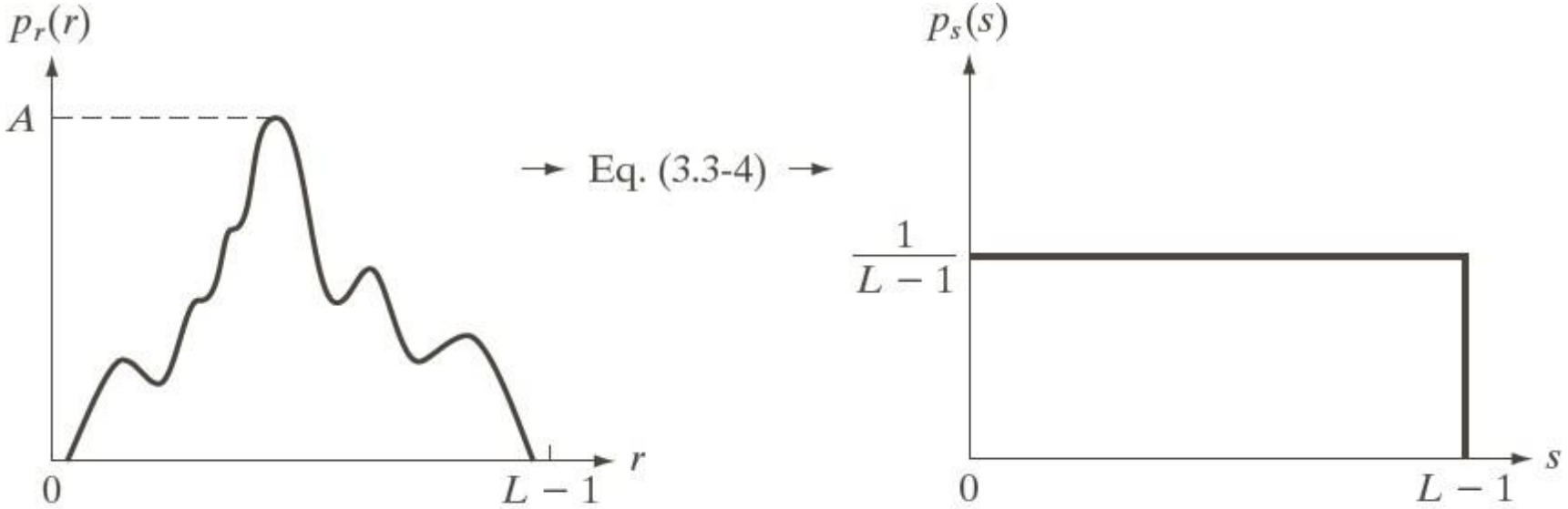
$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L-1)\frac{d}{dr}\left[\int_0^r p_r(w)dw\right] = (L-1)p_r(r)$$

Histogram Equalization

$$p_s(s) = \frac{1}{L-1}$$

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k} \frac{n_j}{MN}, \quad k = 0,1,2,\ldots,L-1$$



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r. The resulting intensities, s, have a uniform PDF, independently of the form of the PDF of the r's.

Histogram Equalization

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1 Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

Histogram Equalization

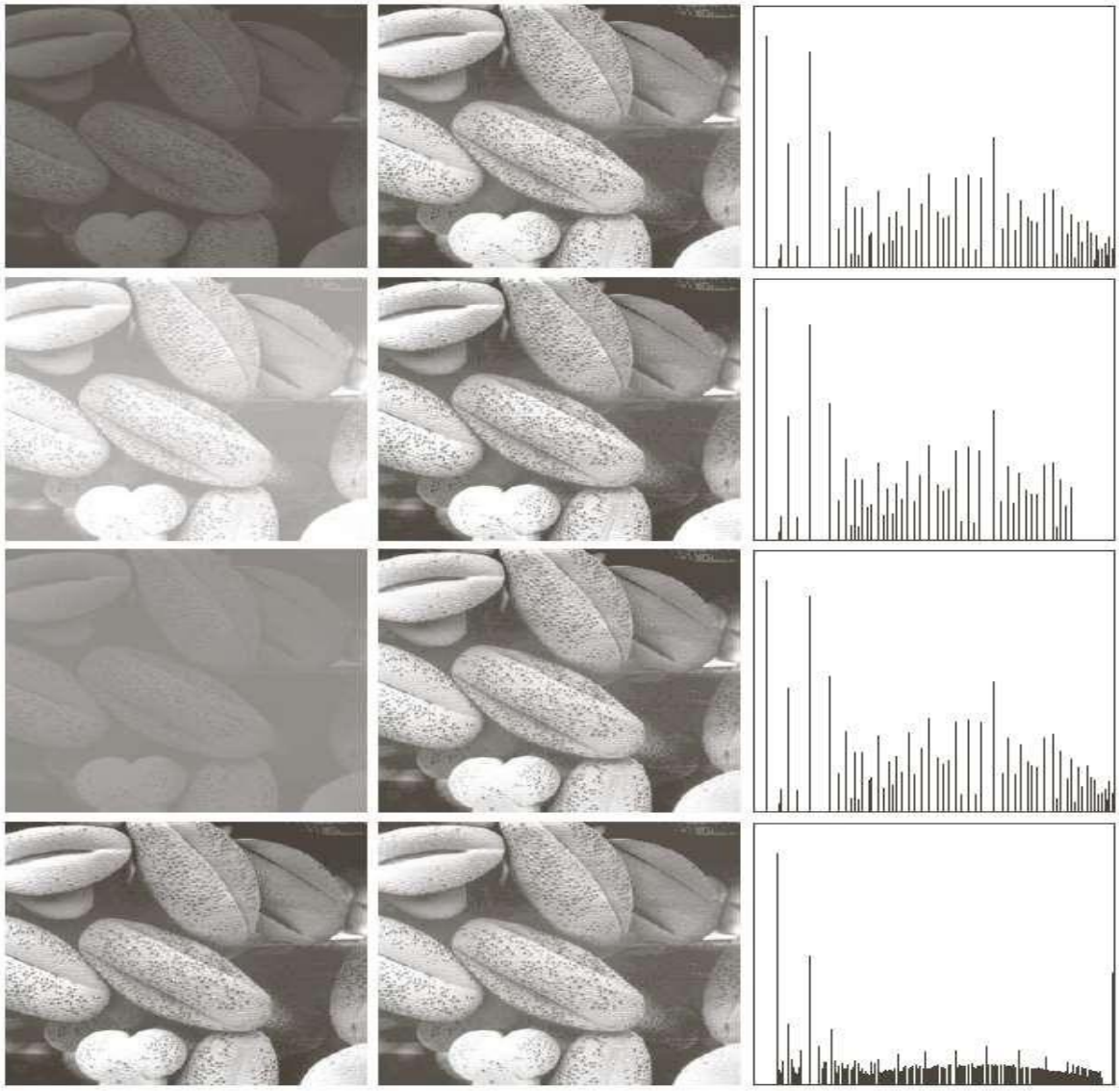


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

Mechanics of Spatial Filtering

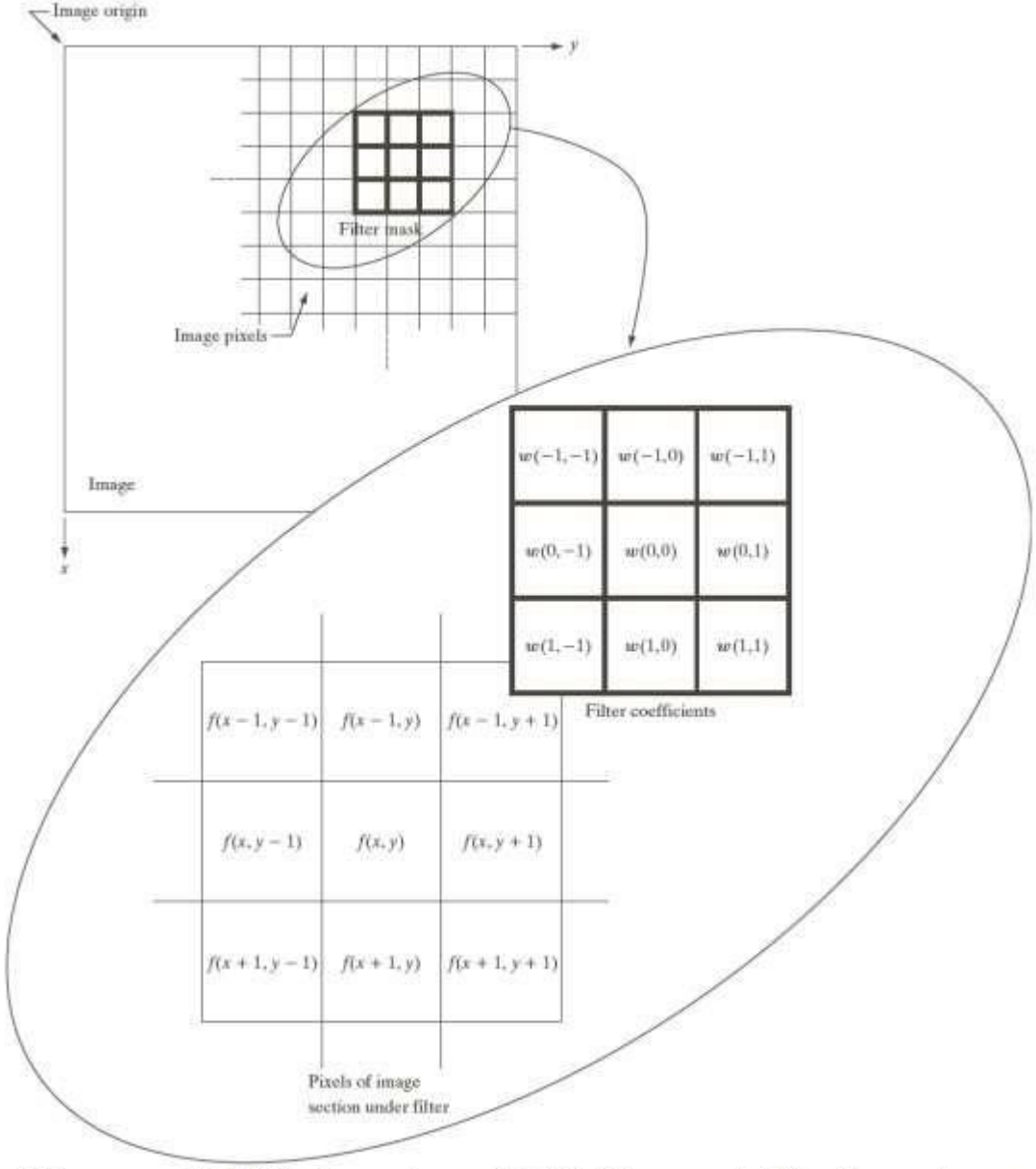


FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

Mechanics of Spatial Filtering

- The result (or response), R, of linear filtering with the filter mask at a point (x,y) in the image is:
- $R = w(-1,-1)f(x-1,y-1) + w(-1,0)f(x-1,y) + \cdots + w(0,0)f(x,y) + \cdots + w(1,0)f(x-1,y-1) + w(1,1)f(x+1,y+1)$

Smoothing Spatial Linear Filters

- The output of a smoothing linear spatial filter is simply the average of the pixels contained in the neighborhood of the linear mask.
- By replacing the value of every pixel in an image by the average of the intensity levels in the neighborhood defined by the filter mask.
- Also called *Lowpass filter.*
- Random noise typically consists of sharp transition.
- Reduce “sharp” transition in intensities.
- If all coefficients are equal in filter then it is also called *a box filter*.

Smoothing Linear Filter

$\frac{1}{9} \times$

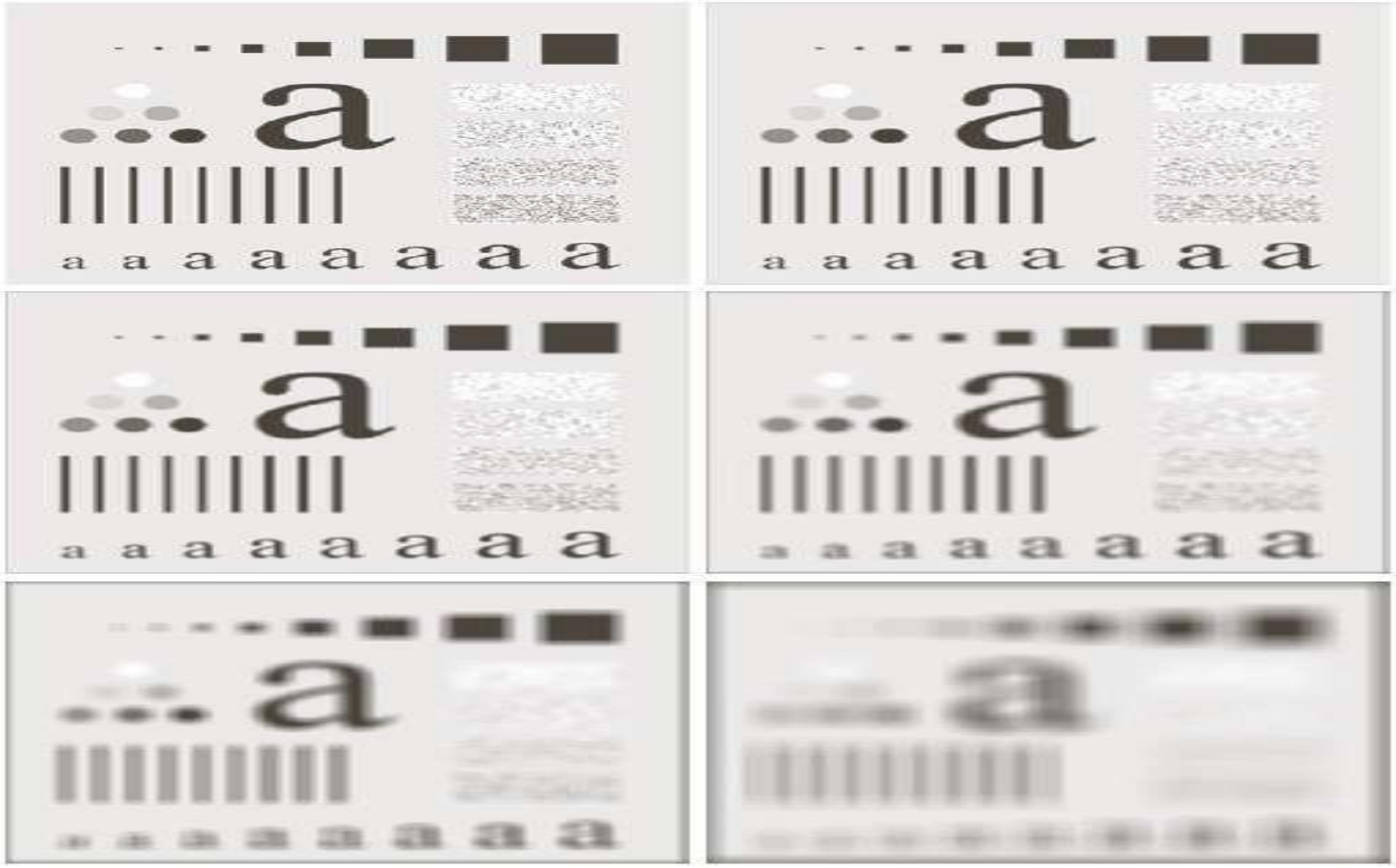
1	1	1
1	1	1
1	1	1

$\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.



a b
c d
e f

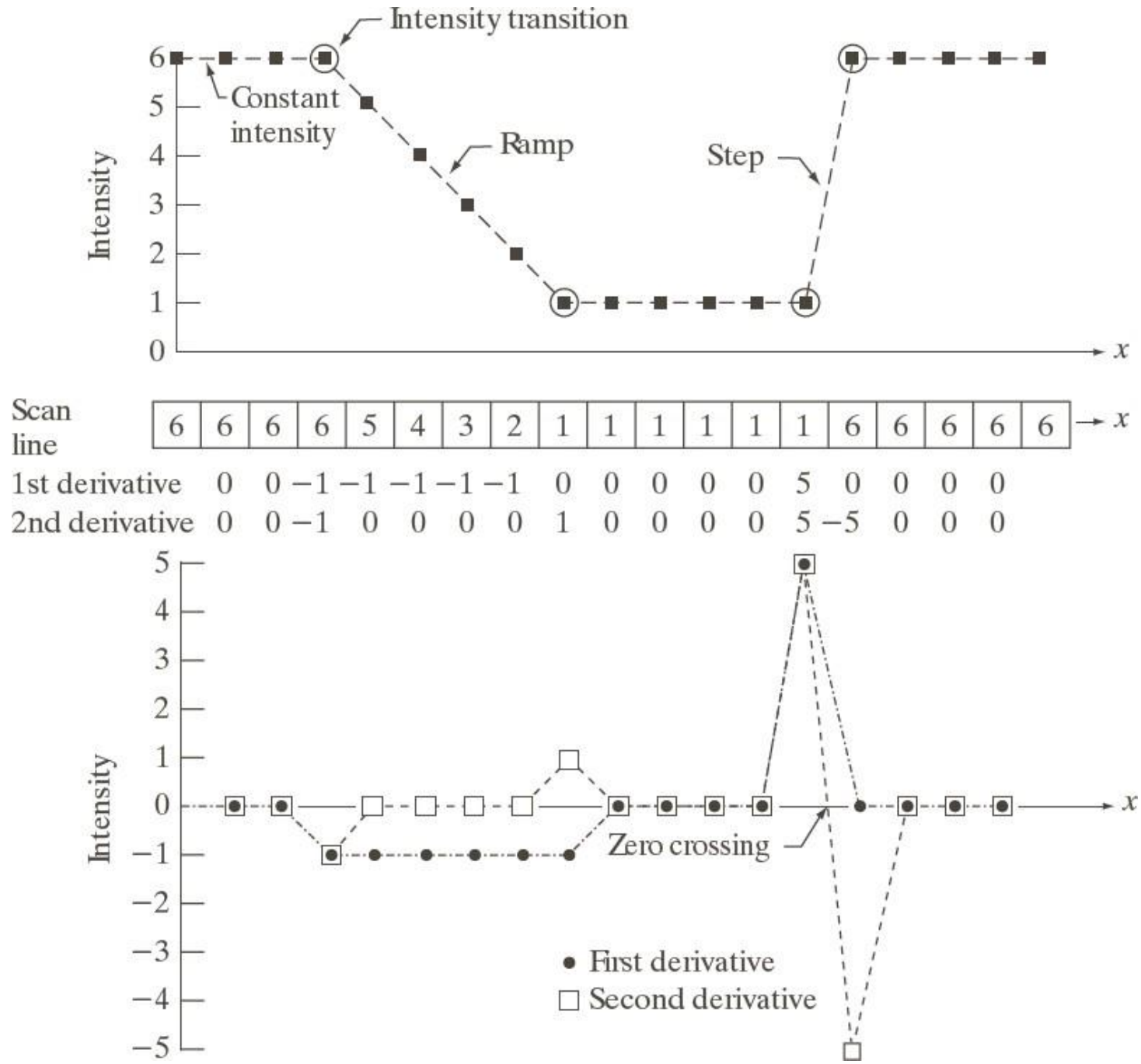
FIGURE 3.33 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35, respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50 × 120 pixels.

Sharpening Spatial Filters

- Objective of sharpening is to highlight transitions in intensity.
- Uses in printing and medical imaging to industrial inspection and autonomous guidance in military systems.
- Averaging is analogous to integration, so sharpening is analogous to spatial differentiation.
- Edges also characterized by sharp intensity transitions, so averaging filters have the undesirable side effect that they blur edge.
- Thus, image differentiation enhances edges and other discontinuities (such as noise) and deemphasizes areas with slowly varying intensities.

Foundation

- Definition for a first order derivative (1) must be zero in areas of constant intensity (2) must be nonzero at the onset of an intensity step or ramp and (3) must be nonzero along ramps.
- For a second order derivatives (1) must be zero in constant areas (2) must be nonzero at the onset and (3) must be zero along ramps of constant slope.
- First order derivative of a one dimensional function f(x) is the difference of f(x+1) – f(x).
- Second order = f(x+1) + f(x-1) -2f(x)



a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

Second Derivatives-The Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial^2 x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y).$$

Second Derivatives - The Laplacian

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

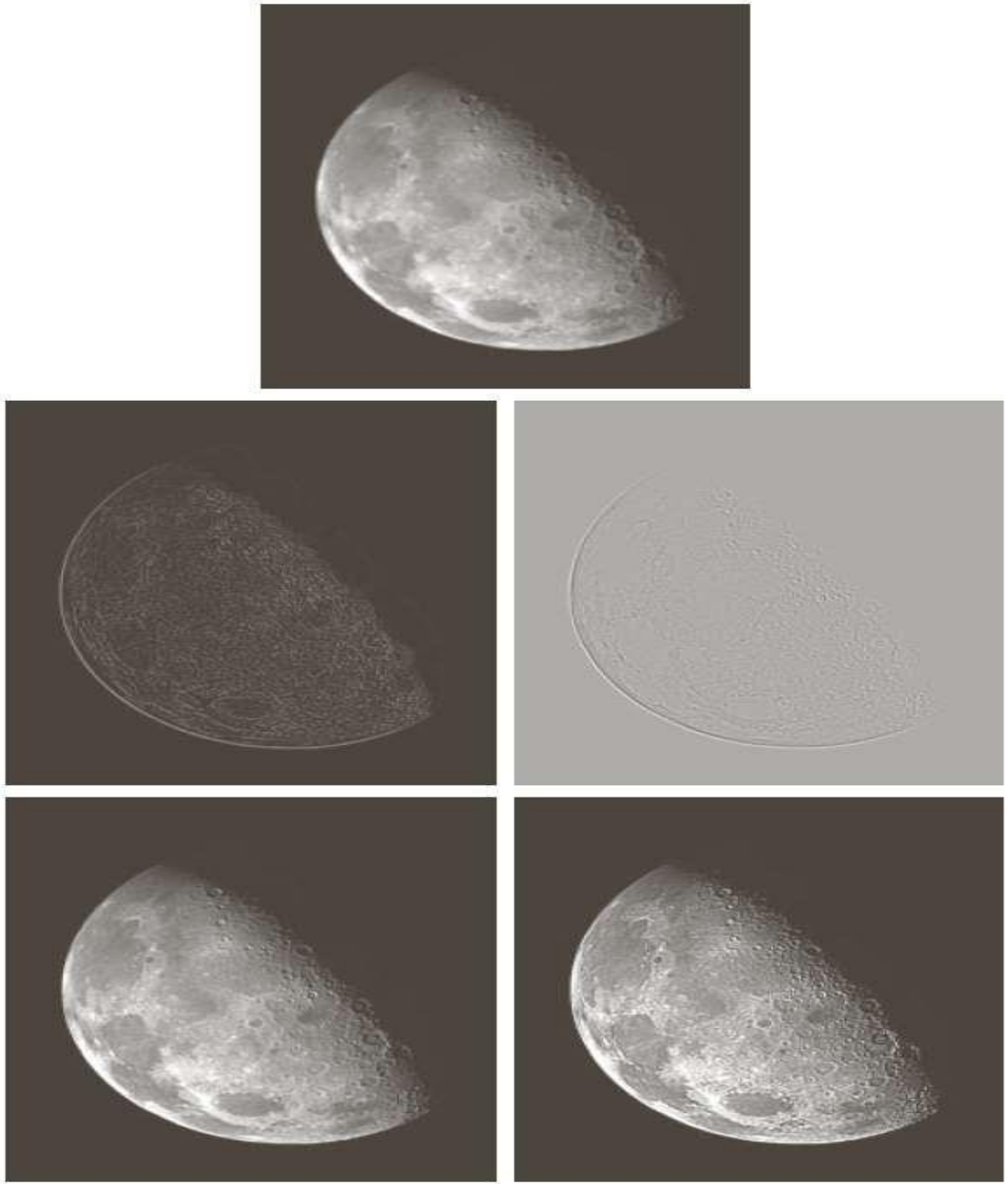
−1	−1	−1
−1	8	−1
−1	−1	−1

a b
c d

FIGURE 3.37
(a) Filter mask used to implement Eq. (3.6-6). (b) Mask used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other implementations of the Laplacian found frequently in practice.

Second Derivatives-The Laplacian

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the Laplacian mask is positive.} \end{cases}$$



a
b c
d e

FIGURE 3.38
(a) Blurred image of the North Pole of the moon. (b) Laplacian without scaling. (c) Laplacian with scaling. (d) Image sharpened using the mask in Fig. 3.37(a). (e) Result of using the mask in Fig. 3.37(b). (Original image courtesy of NASA.)

Unsharp Masking and High boost Filtering

- Unsharp Masking
  - Read Original Image f(x,y)
  - Blurred original image f´(x,y)
  - Mask = f(x,y) – f´(x,y)
  - $f_s(x,y) = f(x,y) + Mask$
  - Use publishing industry to sharpen image
- High Boost Filtering
  - Read Original Image f(x,y)
  - Blurred original image f´(x,y)
  - $f_{hb}(x,y) = Af(x,y) – f´(x,y)$, where A>1
  - Use when input image is darker than desired

First Derivative – The Gradient

$$\nabla \mathbf{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

$$\begin{aligned} \nabla f &= \operatorname{mag}(\nabla \mathbf{f}) \\ &= \left[G_x^2 + G_y^2\right]^{1/2} \\ &= \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{1/2} \end{aligned}$$

$$\nabla f \approx |G_x| + |G_y|$$

First Derivative – The Gradient

- The center point f(x,y) denotes z_5, f(x-1,y-1) denotes z_1 and so on.
- First-order derivative approximates:
  - $G_x = (z_8 - z_5)$ and $G_y = (z_6 - z_5)$
  - $\nabla f \approx |G_x| + |G_y| = |(z_8 - z_5)| + |(z_6 - z_5)|$

Other Masks

- Robert cross-gradient operators:
  - $G_x = (z_9 - z_5)$ and $G_y = (z_8 - z_6)$
  - $\nabla f \approx |G_x| + |G_y| = |(z_9 - z_5)| + |(z_8 - z_6)|$
- Sobel operators:
  - $G_x = (z_7 + 2z_8 - z_9) - (z_1 + 2z_2 - z_3)$
  - $G_y = (z_3 + 2z_6 - z_9) - (z_1 + 2z_4 - z_7)$
  - $|G_x| + |G_y| = |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

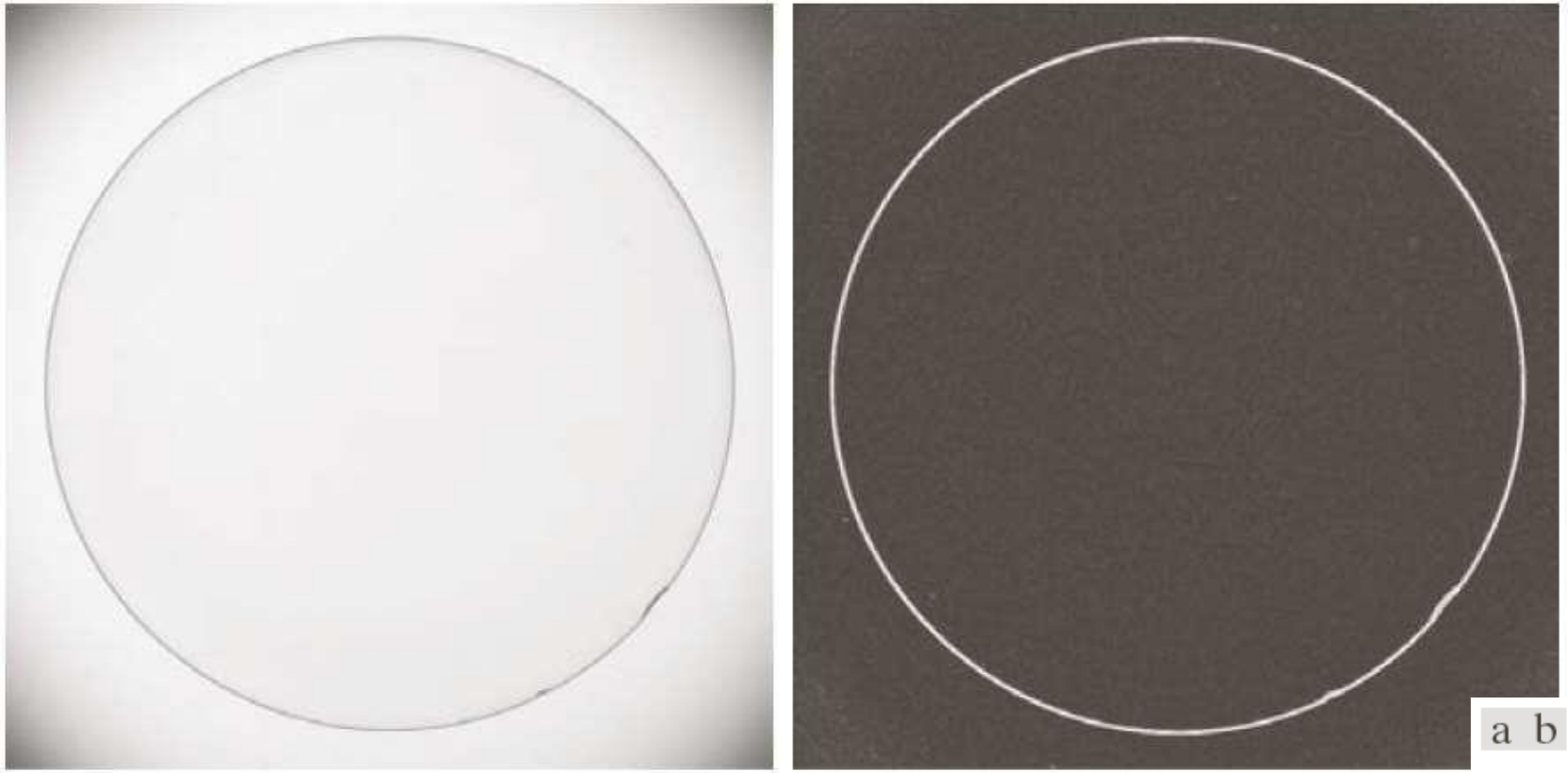
0	−1
1	0

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

a
b c
d e

FIGURE 3.41
A 3 × 3 region of an image (the zs are intensity values).
(b)–(c) Roberts cross gradient operators.
(d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.



a b

FIGURE 3.42
(a) Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Pete Sites, Perceptics Corporation.)